

NBA GAME PREDICTION · WIDE & DEEP NEURAL NET

Can a model hang with Vegas?

A leakage-free Wide & Deep network predicts NBA games from pre-tip-off box-score form — and lands within a few points of the sharpest betting markets on earth.

63%

THIS MODEL

VS 66–68%

LAS VEGAS

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Is 63% good? Depends what you stack it against.

63%

THIS MODEL — held-out test seasons

Pre-game team form only. No injuries, no lineups, no betting signal. Trained on 2015–21, judged on games it had never seen.

66–68%

LAS VEGAS — closing market line

Rooms full of quants, insider injury news, sharp money moving billions. The most accurate public NBA forecast that exists.

A few points apart — that gap is the whole story.

Single-game prediction is deceptively hard

54.4%

Home-win base rate

The naive floor: just pick the home team.

~67%

Las Vegas ceiling

The practical upper bound on public forecasts.

1 game

All variance, no mercy

Shot luck, foul trouble, late-game noise
— unforecastable.

A useful model has to climb from the 54% floor toward the 67% ceiling — and resist the temptation to cheat past it.

Ten seasons, rebuilt to respect time

11,876

games

64

leakage-free features

10

seasons (2015–25)

13.5k

model parameters

1

NBA Stats API

LeagueGameFinder, 10 seasons

>

2

One row / game

home vs away, from MATCHUP

>

3

Rolling L5 & L10 form

shift(1) — no peeking ahead

>

4

Time-based split

train · validation · test by season

An 87% model was lying. Fixing it gave 63%.

87–88%

LEAKED

Two features secretly encoded the result:

- cumulative season win% — a proxy for final standings
- head-to-head win% with $|r| = 0.83$ to the outcome



62.9%

HONEST

Rebuilt time-safe:

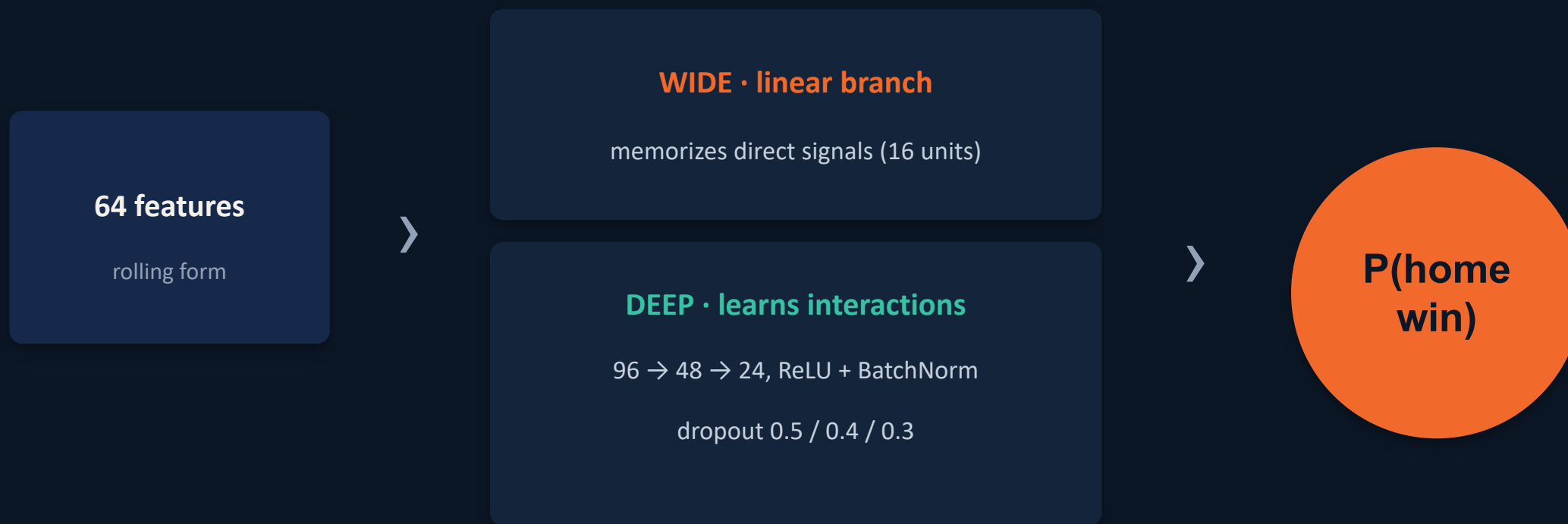
- season win% removed; bounded L5/L10 windows only
- H2H recomputed as a shifted mean $\rightarrow |r| = 0.08$
- audit: every feature $|r| < 0.40$

The lesson

On temporal data, a score that looks too good is a symptom — not a success.

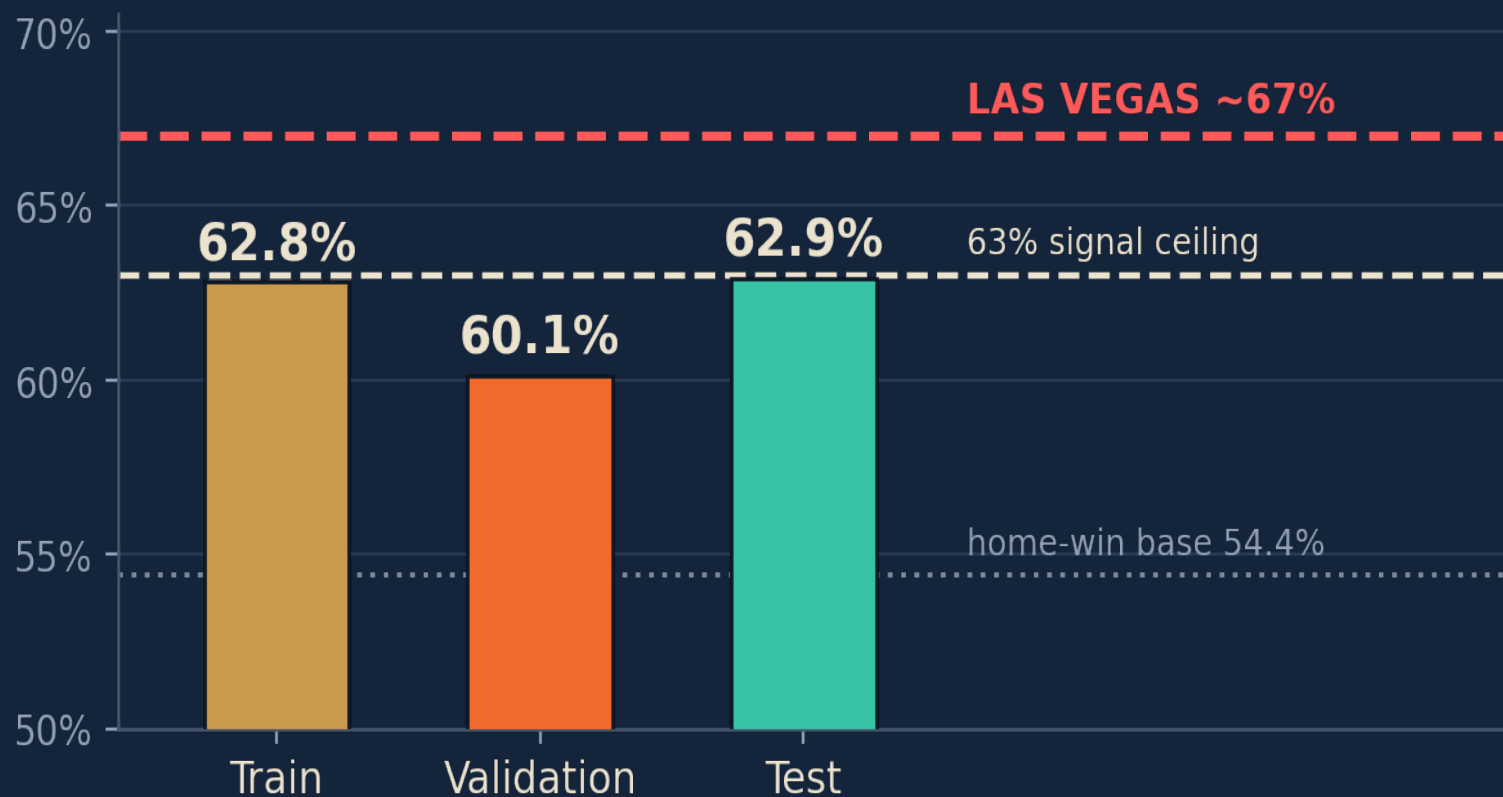
The right reaction to 90% here is suspicion: which feature is encoding the answer?

Wide & Deep: memorize and generalize



~13,500 parameters · Adam, weight-decay 2e-3 · early stopping (best @ epoch 6) · heavy dropout keeps it pinned at the ceiling, not past it

Every split lands at the ceiling — just shy of Vegas



62.9%

Test acc

0.676

Test ROC-AUC

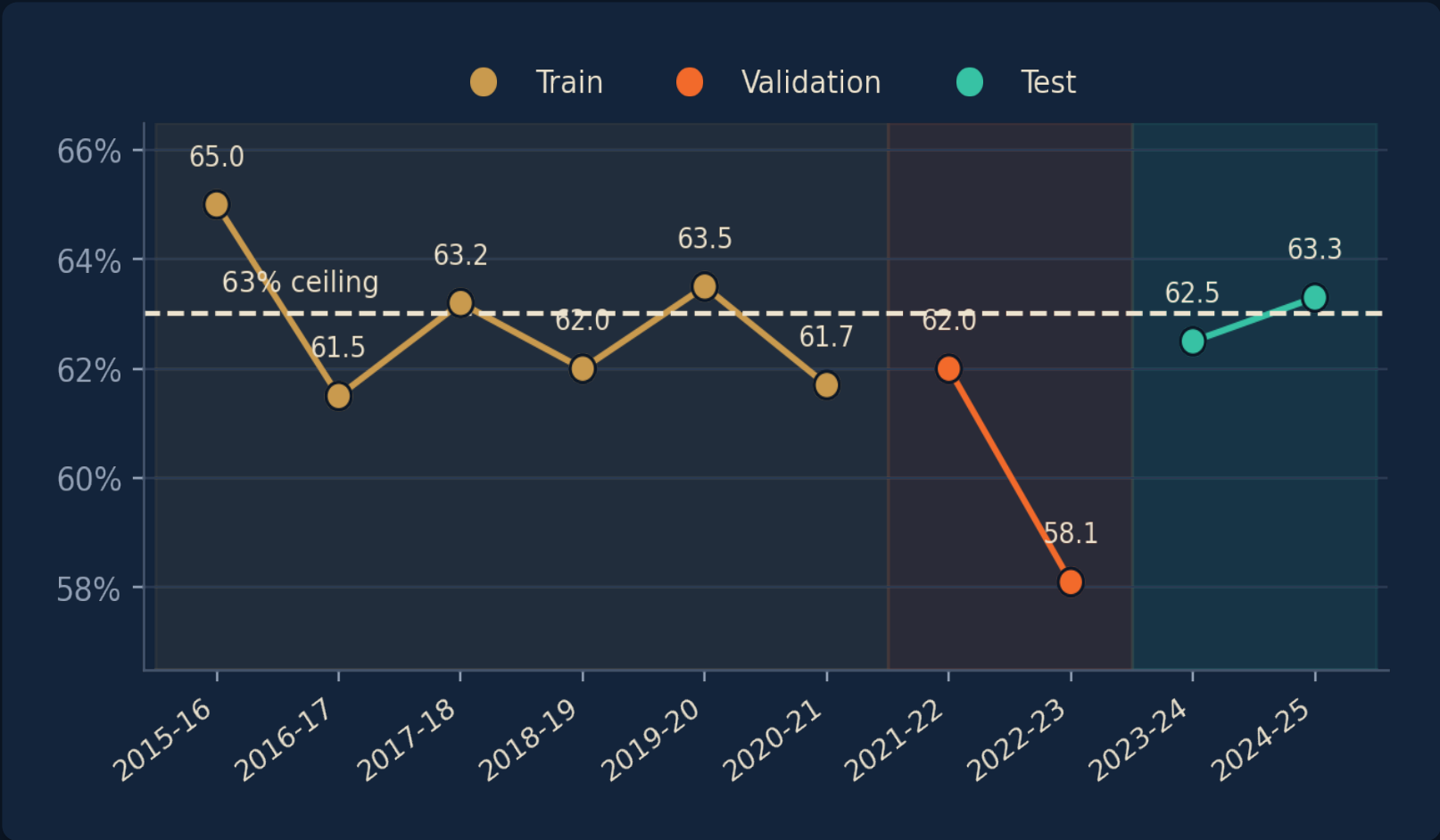
62.8%

Train acc

~4 pts

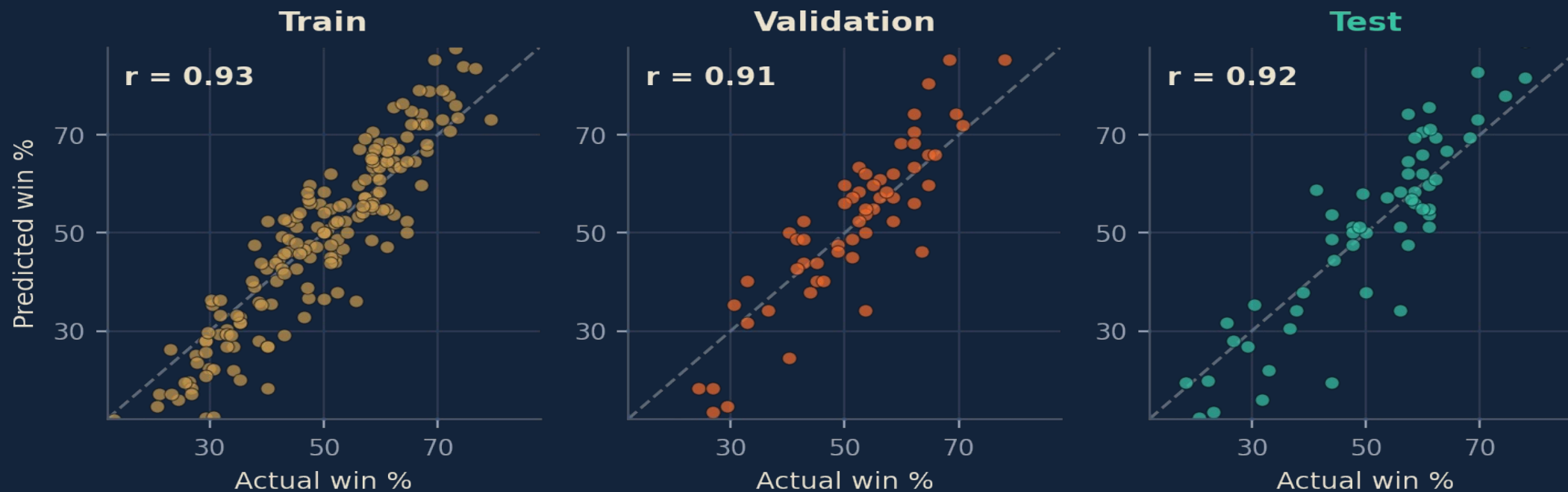
Gap to Vegas

Accuracy holds season to season



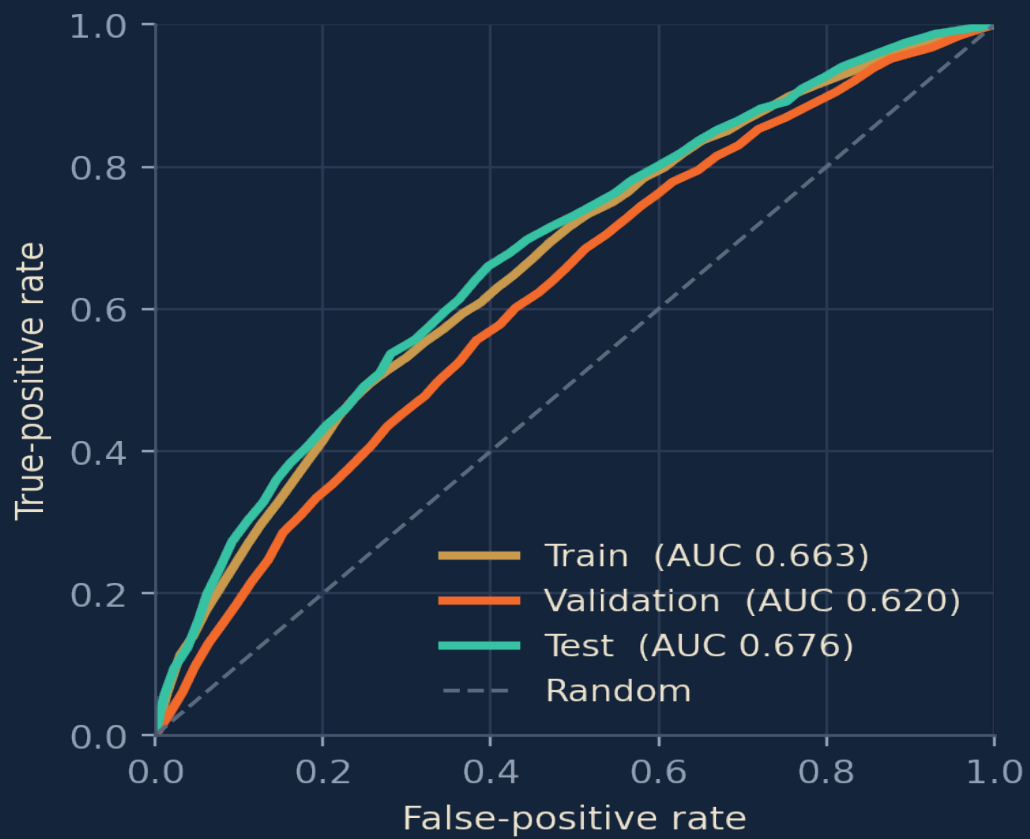
Train ≈ Test
The held-out 2023–24 and 2024–25 seasons score right alongside the training seasons. The model generalizes to years it never saw rather than memorizing the past.

Predicted win% tracks reality



Each point is one team-season. Hugging the diagonal means the model's projected record matches what teams actually did — and it holds on the unseen test seasons ($r = 0.92$).

Well above chance in every split



Test AUC 0.676

Best of the three splits — performance does not decay on unseen data.

Train AUC 0.663

Close to test: the model is not memorizing.

Validation AUC 0.620

The noisy 2022–23 season pulls this down slightly.

Three reasons 63% is exactly right

THE WALL

63% is the real limit

Bigger, deeper, less-regularized models did not move the number. That's the signal ceiling of the data, not a tuning failure.

THE LEAK TEST

A high score is a red flag

If this returned 90%, the right response is suspicion. Honest features cap out near the market — anything above means leakage.

THE FIT

Train $\approx$ Test

62.8% train vs 62.9% test, within a tenth of a point. The model generalizes; dropout and weight decay keep it honest.

THE ANSWER

Yes — a model can hang with Vegas.

A leakage-free Wide & Deep network reaches 62.9% on unseen seasons — a few points under the ~67% market line, with the gap fully explained by the injury and lineup information Vegas prices in and the model deliberately omits.

- › **Real contribution** Diagnosing and removing temporal leakage — not chasing a vanity number.
- › **Honest by design** 62.9% that survives deployment beats a brittle 90% that collapses.
- › **Next** Sequence models, injury & lineup data, and calibration against closing lines.